

Midterm 2 Take Home Paper

YOUR NAME

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```
library(tidyverse)
library(openintro)
library(infer)
```

Directions: In this project, you will apply various statistical procedures covered in our course. If you have not done so before, make sure to load the necessary R packages, including `openintro` and `tidyverse`, by running the `library()` functions given above.

The code chunk below will help you load your data. You do not have to use all of the datasets. Examine each of them and feel free to choose one based on your interest. Remember that some datasets can only be used for comparing proportions or conducting a paired sample t -test.

```
data("yawn")
data("possum")
data("lizard_run")
data("lego_population")
data("hsb2")
data("fheights")
data("exam_grades")
data("biontech_adolescents")
```

To see the details related to these datasets, examine the links below. Use the `.txt` files when using the applets.

- [yawn](#)
- [possum](#)
- [lizard_run](#)
- [lego_population](#)
- [hsb2](#) - High School and Beyond Survey
- [fheights](#) - Female Heights
- [exam_grades](#)
- [biontech_adolescents](#)

Part 1: Getting to Know Your Dataset

Question 1.1: Use the `glimpse()` function to introduce the dataset(s) you will use for this project.

```
# Type your code here
```

Type your answer here.

Question 1.2: Identify the types of variables in the dataset (categorical, binary categorical, or quantitative).

Type your answer here.

Question 1.3: Choose a quantitative variable and visualize its distribution. Describe the graph.

```
# Type your code here
```

Type your answer here.

Question 1.4: Choose a categorical variable and create a visualization. Describe the graph.

```
# Type your code here
```

Type your answer here.

Question 1.5: Examine the dataset carefully. Do you see any potential outliers? How would you handle them if you do?

Type your answer here.

Part 2: Statistical Inference

In this section, you will perform inferential statistical tests using your dataset.

One Sample t -test

In this section, you are expected to write a case study based on your dataset. Choose a quantitative variable and conduct a one-sample t -test. Show your work.

Type your case study here.

Question 2.1: What is the research question of this study?

Type your answer here.

Question 2.2: What type of variable do we have in this study?

Type your answer here.

Question 2.3: List the assumptions required for this test and check the assumption of normality for this study. Interpret the assumptions overall.

- **Random Sampling:** Explain here.
- **Independence of Observations:** Explain here.
- **Normal Distribution:** Explain here.

```
# Type your code here to check normality
```

Interpret your output here.

Question 2.4: Write out the null and alternative hypotheses in words, in the context of this study.

Type your answer here.

Question 2.5: Compute the t -statistic and explain what you see in the output.

```
# Type your code here
```

Type your answer here.

Question 2.6: Draw a conclusion and write a conclusion statement. Provide the context clearly.

Type your answer here.

Independent Samples t -test

In this section, you are expected to write a case study based on your dataset. Choose one quantitative variable and one categorical variable (with two categories) and conduct an independent samples t -test. Show your work.

Question 3.1: What is the research question of this study?

Type your answer here.

Question 3.2: What type of variables do we have in this study?

Type your answer here.

Question 3.3: List the assumptions required for this test and check the assumption of normality for this study. Interpret the assumptions overall.

- **Random Sampling:** Explain here.
- **Independence of Observations:** Explain here.
- **Normal Distribution:** Explain here.

```
# Type your code here to check normality
```

Interpret the assumptions and explain your reasoning.

Question 3.4: Write out the null and alternative hypotheses in words, in the context of this study.

Type your answer here.

Question 3.5: Compute the t -statistic and explain what you see in the output.

```
# Type your code here
```

Type your answer here.

Question 3.6: Draw a conclusion and write a conclusion statement. Provide the context clearly.

Type your answer here.

Paired Sample *t*-test

Write a case study based on your dataset. Show your work.

Question 4.1: What is the research question of this study?

Type your answer here.

Question 4.2: What type of variable do we have in this study?

Type your answer here.

Question 4.3: List the assumptions required for this test and check the assumption of normality for this study. Interpret the assumptions overall.

List and explain the assumptions here.

```
# Type your code here to check normality
```

Interpret the assumptions and explain your reasoning.

Question 4.4: Write out the null and alternative hypotheses in words, in the context of this study.

Type your answer here.

Question 4.5: Compute the *t*-statistic and explain what you see in the output.

```
# Type your code here
```

Type your answer here.

Question 4.6: Draw a conclusion and write a conclusion statement. Provide the context clearly.

Type your answer here.

Comparing Two Proportions

Is yawning contagious? Conventional wisdom says yes: when you see someone else yawn, you're prone to feel sleepy and let out a yawn yourself. How many times have you caught yourself in this situation, or noticed it in someone else? But will this hypothesis withstand a scientific test? Will data support this claim?

The folks at *MythBusters*, a popular television program on the Discovery Channel, investigated this issue by using a two-way mirror and a hidden camera. Fifty subjects sat in a booth, accompanied only by an experimental attendee. For some of the subjects, the attendee yawned (planting a yawn "seed"), while for other subjects the attendee did not yawn. The researchers decided in advance, with a random mechanism, which subjects would receive the yawn seed and which would not. As time passed, the researchers watched to see which subjects yawned.

Use the [applet](#) and select the dataset "**Yawning**". Complete this part by using the applet.

Question 5.1: What is the research question of this study?

Type your answer here.

Question 5.2: What type of variable do we have in this study?

Type your answer here.

Question 5.3: Write out the null and alternative hypotheses in words, in the context of this study.

Type your answer here.

Question 5.4: Let's use simulation-based approach to see if there is any association between these two variables. Report your simulation-based p value.

Type your answer here.

Question 5.6: Draw a conclusion and write a conclusion statement. Provide the context clearly.

Type your answer here.